

# Global Angular Blocks for the Middle-Ratio Shell Proxy

Global analytic angular module

## Abstract

The global structural reduction for the spherical-shell proxy leaves an angular rounding defect  $\mathcal{E}_{\text{ang}}$  and a smooth positive payment  $\mathcal{H}$ . This note controls the defect for every  $0 < \rho < 1$  and  $K > 0$  with at least one active radial layer. A single paired radial-block argument, using only the sharp slope cap  $-A' \leq 1/2$  and the exact strict angular rounding, gives

$$\mathcal{E}_{\text{ang}} < \frac{(1 - \rho^2)K^2}{6} - \frac{N}{2},$$

where  $N$  is the number of active shifted radial layers. On the upward-closed middle-ratio region  $\rho_c \leq \rho \leq 39/50$ ,  $K \geq k_{11}(\rho)$ , one has  $N \geq 1$ . The global middle-ratio theorem is therefore reduced to the smooth scalar inequality (19), which is established in the companion scalar-positivity module.

## 1 Definitions and the target payment

For  $a > 0$  and  $z \geq 0$ , define

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left( \sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \quad (1)$$

Fix  $0 < \rho < 1$  and  $K > 0$ , and put

$$A(z) := G_K(z) - G_{\rho K}(z), \quad T := A(0) = \frac{(1 - \rho)K}{\pi}. \quad (2)$$

The function  $A$  decreases strictly from  $T$  to 0. Denote its decreasing inverse by  $R : [0, T] \rightarrow [0, K]$ , and set

$$F(t) := R(t)^2, \quad q(z) := -A'(z), \quad \tau := A(\rho K) = KG_1(\rho). \quad (3)$$

Direct differentiation gives

$$q(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \frac{z}{K}, & \rho K \leq z < K. \end{cases} \quad (4)$$

Consequently

$$0 \leq q(z) \leq \frac{1}{2}, \quad q \text{ increases on } (0, \rho K), \quad q \text{ decreases on } (\rho K, K). \quad (5)$$

Let

$$x_n := n - \frac{1}{4}, \quad M(r) := \max \left\{ 0, \left\lceil r - \frac{1}{2} \right\rceil \right\}, \quad N := \#\{n \geq 1 : x_n < T\}. \quad (6)$$

For  $1 \leq n \leq N$ , write

$$r_n := R(x_n), \quad m_n := M(r_n), \quad (7)$$

and define the exact angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{n=1}^N (m_n^2 - r_n^2). \quad (8)$$

The strict convention is built into  $M$ : at  $r = m + 1/2$  one has  $M(r) = m$ , not  $m + 1$ .

The smooth payment obtained from the retained radial Stieltjes remainder is

$$\begin{aligned} \mathcal{H}(\rho, K) := & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi \rho K}{\arccos \rho} \left( \frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz. \end{aligned} \quad (9)$$

The preceding structural reduction proves the direct lower bound

$$W(\rho, K) - P(\rho, K) \geq \mathcal{H}(\rho, K) - \mathcal{E}_{\text{ang}}(\rho, K). \quad (10)$$

This note derives a smooth upper bound for the left side.

## 2 A one-layer paired-block estimate

The cap  $q \leq 1/2$  in (5) forces a useful separation: one unit of radial action spans at least two units of angular radius. The shift  $x_n = n - 1/4$  leaves a left block of length  $3/4$ , enough to pay the worst angular rounding in that layer.

**Lemma 1** (One-layer paired block). *For every active  $n$ ,*

$$r_n \leq \frac{4}{3} \int_{n-1}^{x_n} R(t) dt - \frac{3}{4}, \quad (11)$$

and

$$m_n^2 - r_n^2 < r_n + \frac{1}{4}. \quad (12)$$

*Both statements remain valid, with the displayed strictness, at every radial-angular common wall.*

*Proof.* Fix  $t \in [n-1, x_n]$  and put  $z = R(t) \geq r_n$ . Since  $A(r_n) = x_n$ , (5) gives

$$x_n - t = A(r_n) - A(z) = \int_{r_n}^z q(s) ds \leq \frac{z - r_n}{2}.$$

Hence

$$R(t) \geq r_n + 2(x_n - t).$$

Integration over the interval of length  $3/4$  yields

$$\int_{n-1}^{x_n} R(t) dt \geq \frac{3}{4}r_n + \frac{9}{16},$$

which is (11).

By strict counting,  $m_n < r_n + 1/2$  even when  $r_n$  is a half-integer: at  $r_n = m + 1/2$  the value is  $m_n = m$ . Squaring gives

$$m_n^2 < (r_n + 1/2)^2 = r_n^2 + r_n + 1/4,$$

and proves (12).  $\square$

**Proposition 2** (Global angular block bound). *For every  $0 < \rho < 1$  and every  $K > 0$  with  $N \geq 1$ ,*

$$\boxed{\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{N}{2}.} \quad (13)$$

*In particular,*

$$\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{1}{2}. \quad (14)$$

*Proof.* The left blocks  $[n - 1, x_n]$ ,  $1 \leq n \leq N$ , are disjoint subsets of  $[0, T]$ . Summing (11) and then (12) gives

$$\begin{aligned} \mathcal{E}_{\text{ang}} &< \sum_{n=1}^N \left( r_n + \frac{1}{4} \right) \\ &\leq \frac{4}{3} \sum_{n=1}^N \int_{n-1}^{x_n} R(t) dt - \frac{N}{2} \\ &\leq \frac{4}{3} \int_0^T R(t) dt - \frac{N}{2}. \end{aligned}$$

The area identity for decreasing inverses gives

$$\int_0^T R(t) dt = \int_0^K A(z) dz.$$

Scaling  $z = as$  in (1) and using

$$\int_0^1 \sqrt{1 - s^2} ds = \frac{\pi}{4}, \quad \int_0^1 s \arccos s ds = \frac{\pi}{8},$$

shows that

$$\int_0^a G_a(z) dz = \frac{a^2}{8}.$$

Consequently

$$\int_0^T R(t) dt = \frac{(1 - \rho^2)K^2}{8},$$

which proves (13). Equation (14) follows from  $N \geq 1$ .  $\square$

Combining the structural and angular estimates removes the staircase defect from the live handoff altogether.

**Proposition 3** (Direct analytic proxy gap). *For every  $0 < \rho < 1$  and  $K > 0$  with  $N \geq 1$ ,*

$$\boxed{W(\rho, K) - P(\rho, K) > \mathcal{H}(\rho, K) - \frac{(1 - \rho^2)K^2}{6} + \frac{N}{2}.} \quad (15)$$

*This one inequality includes the first and terminal radial pieces and every strict angular wall; no separate common-jump estimate is a premise.*

*Proof.* Insert (13) into (10). □

### 3 The upward-closed middle-ratio scalar inequality

Set

$$\rho_c := \frac{1}{1 + 2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1 - \rho)^2} + 132}, \quad (16)$$

and consider the upward-closed middle-ratio region

$$\mathcal{R}_c := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho) \right\}. \quad (17)$$

On this set,

$$T \geq \frac{(1 - \rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1 - \rho)^2}{\pi^2}} > 1, \quad (18)$$

so  $N \geq 1$ . Proposition 3 reduces the global middle-ratio theorem to the single inequality

$$\boxed{\mathcal{S}(\rho, K) := \mathcal{H}(\rho, K) - \frac{(1 - \rho^2)K^2}{6} + \frac{1}{2} > 0} \quad ((\rho, K) \in \mathcal{R}_c). \quad (19)$$

Everything in  $\mathcal{S}$  is smooth at every finite point of  $\mathcal{R}_c$ ; the floor functions and all radial–angular common jumps have disappeared. Substituting (9) gives the explicit form

$$\begin{aligned} \mathcal{S}(\rho, K) &= \frac{(5\rho^2 - 2)K^2}{12} - \frac{\pi\rho K}{4(1 - \rho)} + \frac{1}{2} \\ &\quad + \frac{2\pi\rho K}{\arccos \rho} \left( \frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32} \right) \\ &\quad + \int_{\rho K}^K 2z \frac{A(z)(1 + A(z))}{(1 + 2A(z))^2} dz. \end{aligned} \quad (20)$$

The elementary identity

$$\frac{u(1 + u)}{(1 + 2u)^2} = \frac{1}{4} - \frac{1}{4(1 + 2u)^2} \quad (21)$$

turns the same scalar gap into the more useful loss form

$$\begin{aligned} \mathcal{S}(\rho, K) &= \frac{(1 + 2\rho^2)K^2}{12} - \frac{1}{2} \int_{\rho K}^K \frac{z}{(1 + 2A(z))^2} dz \\ &\quad - \frac{\pi\rho K}{4(1 - \rho)} + \frac{1}{2} + \frac{2\pi\rho K}{\arccos \rho} \left( \frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32} \right). \end{aligned} \quad (22)$$

Thus the only non-endpoint quantity still requiring control is the positive boundary-layer loss integral in (22).

*Remark 4* (Status and next step). The direct proxy gap (15) is proved analytically from one retained-radial estimate and one one-layer angular estimate. The first remaining step within this module is the smooth scalar inequality (19). The companion scalar-positivity module proves it on all of  $\mathcal{R}_c$  by a uniform upper bound for the boundary-layer loss in (22) and strict positivity on the base curve. No upper cutoff in  $K$  occurs in the angular estimate or in the verification that  $N \geq 1$ .